

LIST OF PROBLEMS ABOUT BELIEFS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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1. Consider the following belief space with a common prior:

$$\begin{aligned} N &= \{1, 2\}, \quad \Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\} \\ \mathcal{P}_1 &= \{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}\}, \quad \mathcal{P}_2 = \{\{\omega_1, \omega_2, \omega_3\}, \{\omega_4\}\} \\ \mu(\omega_1) &= \mu(\omega_2) = \mu(\omega_3) = \mu(\omega_4) = \frac{1}{4}. \end{aligned}$$

Let $A = \{\omega_1, \omega_3\}$.

- Show that at every state of the world, it is common knowledge that player 1 ascribes probability 1/2 to the event A .
- Show that in states ω_1, ω_2 , and ω_3 , player 2 ascribes probability 2/3 to the event A , and that in state of the world ω_4 , she ascribes probability 0 to event A .
- Does this example contradict Aumann's Agreement theorem?

2. Consider the following belief space:

$$\begin{aligned} N &= \{1, 2\}, \quad \Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}, \\ \Sigma_1 &= \{\emptyset, \{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}, \Omega\}, \quad \Sigma_2 = \{\emptyset, \{\omega_1, \omega_4\}, \{\omega_2, \omega_3\}, \Omega\}, \\ \mu_1(\omega_1) &= \mu_1(\omega_2) = \mu_1(\omega_3) = \mu_1(\omega_4) = \frac{1}{4}, \\ \mu_2(\omega_1) &= \mu_2(\omega_3) = \frac{2}{10}, \quad \mu_2(\omega_2) = \mu_2(\omega_4) = \frac{3}{10}. \end{aligned}$$

Let $A = \{\omega_1, \omega_3\}$

- Show that it is common knowledge at all states of the world that player 1 ascribes probability 1/2 to the event A .
- Show that it is common knowledge at all states of the world that player 2 ascribes probability 2/5 to the event A .
- Conclude that the players 1 and 2 *agree to disagree* about the probability of event A , and that this disagreement is common knowledge. Does this contradict Aumann's Agreement Theorem?

3. Consider the following belief space with a common prior:

$$\begin{aligned} N &= \{1, 2\}, \quad \Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\} \\ \Sigma_1 &= \{\emptyset, \{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}, \Omega\}, \quad \Sigma_2 = \{\emptyset, \{\omega_1, \omega_2, \omega_3\}, \{\omega_4\}, \Omega\} \\ \mu(\omega_1) &= \mu(\omega_4) = \frac{1}{3}, \quad \mu(\omega_2) = \mu(\omega_3) = \frac{1}{6}. \end{aligned}$$

Let $A = \{\omega_2, \omega_3\}$ and suppose the true state of the world is ω_3 .

- Compute the posterior for player 1: $q_1^0 = \mu_1(A|\Sigma_1)(\omega_3)$.
- Compute the posterior for player 2: $q_2^0 = \mu_2(A|\Sigma_2)(\omega_3)$.
- Suppose player 1 announces q_1^0 . Compute $q_2^1 = \mu_2(A|\Sigma_2, q_1^0)(\omega_3)$.
- Suppose player 2 announces q_2^0 . Compute $q_1^1 = \mu_1(A|\Sigma_1, q_2^0)(\omega_3)$.
- Suppose player 1 announces q_1^1 . Compute $q_2^2 = \mu_2(A|\Sigma_2, q_1^1, q_2^0)(\omega_3)$.
- Suppose player 2 announces q_2^1 . Compute $q_1^2 = \mu_1(A|\Sigma_1, q_2^1, q_2^0)(\omega_3)$.
- What happens to the sequence $(q_1^t, q_2^t)_{t=0,1,\dots}$? Does the sequence converge, or oscillate periodically (or neither)?

Repeat the steps above for the cases in which:

- The true state of the world is ω_1 .

- The true state of the world is ω_2 .

4. Repeat exercise (3.) with the following belief space:

$$N = \{1, 2\}, \quad \Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$\mathcal{P}_1 = \left\{ \{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\} \right\}, \quad \mathcal{P}_2 = \left\{ \{1, 2, 3, 4\}, \{5, 6, 7, 8\}, \{9\} \right\}$$

$$\mu(\omega) = \frac{1}{9}, \quad \forall \omega \in \Omega,$$

and for the event $A = \{1, 5, 9\}$ when the true state of the world is

- $\omega = 9$.
- $\omega = 8$.
- $\omega = 6$.
- $\omega = 4$.
- $\omega = 1$.

5. Repeat exercise (3.) with the following belief space:

$$N = \{1, 2\}, \quad \Omega = \{1, 2, 3, 4, 5\}$$

$$\mathcal{P}_1 = \left\{ \{1, 2\}, \{3, 4\}, \{5\} \right\}, \quad \mathcal{P}_2 = \left\{ \{1, 3, 5\}, \{2\}, \{4\} \right\}$$

$$\mu(\omega) = \frac{1}{5}, \quad \forall \omega \in \Omega,$$

and for the event $A = \{1, 4\}$ and when the true state of the world is $\omega = 4$.

6. Repeat exercise (3.) with the following belief space:

$$N = \{1, 2\}, \quad \Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$\mathcal{P}_1 = \left\{ \{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\} \right\}, \quad \mathcal{P}_2 = \left\{ \{1, 2, 3, 4\}, \{5, 6, 7, 8\}, \{9\} \right\}$$

$$\mu_1(\omega) = \frac{\omega}{45}, \quad \forall \omega \in \Omega, \quad \mu_2(\omega) = \frac{10 - \omega}{45}, \quad \forall \omega \in \Omega$$

and for the event $A = \{1, 5, 9\}$ when the true state of the world is

- $\omega = 9$.
- $\omega = 8$.
- $\omega = 6$.
- $\omega = 4$.
- $\omega = 1$.

Math Corner

7. Prove the law of total expectation (iterated expectations) for a finite set of states of the world. Suppose Ω is finite. Let S be an element of an algebra $\Pi \subseteq 2^\Omega$, and let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Then

$$E[X|S] = E[E[X|\Pi]|S]$$

Hint: Try writing S as a disjoint union of elements in Π .

8. Let $(N, \Omega, \{\mathcal{P}_i\}_{i \in N}, \{\mu_i\}_{i \in N})$ be a belief space described in terms of the players' information partitions. Further, assume there is a common prior $\mu = \mu_i$, $i = 1, \dots, n$ and assume that $\mu(\omega) > 0$ for all $\omega \in \Omega$. Prove the following: For each $\omega \in \Omega$, for each player $i \in N$, and for every $A \subseteq \Omega$, player i knows event A in state of the world ω if and only if she attributes probability 1 to that event:

$$\mu(A|P_i(\omega)) = 1 \iff P_i(\omega) \subseteq A.$$

9. Let $(N, \Omega, \{\mathcal{P}_i\}_{i \in N}, \mu)$ be a belief space with a common prior. Prove that if in state of the world ω Player 1 knows that Player 2 knows A , then $\mu(A|P_1(\omega)) = 1$.